Title: Web Server Deployment and Access Management using AWS EC2

- **Department:** Information Technology
- **Course Code:** 6IT1284
- **Course Name:** Hybrid Cloud Architecture and Management
- **Course Instructor:** Dr Durgadevi P
- **Team Number:** 06
- **Date:** 18-03-2026

Team Members:

1. N Nandish-2023BIFT07AED040
2. Harshith Gowda M- 2023BIFT07AED052
3. Lokesh Naidu CH- 2023BIFT07AED033

Introduction

- Cloud computing allows users to deploy applications and services over the internet instead of local servers.
- Amazon Web Services (AWS) provides **Elastic Compute Cloud (EC2)** to create virtual machines in the cloud.
- A web server can be deployed on EC2 to host websites and web applications.
- Access management ensures that only authorized users can access and manage the server securely.
- This project demonstrates how to deploy a web server on AWS EC2 and configure secure access using cloud security features.

Basic Concept

What it is

Deployment of a web server on a cloud-based virtual machine using **AWS EC2** to host a website.

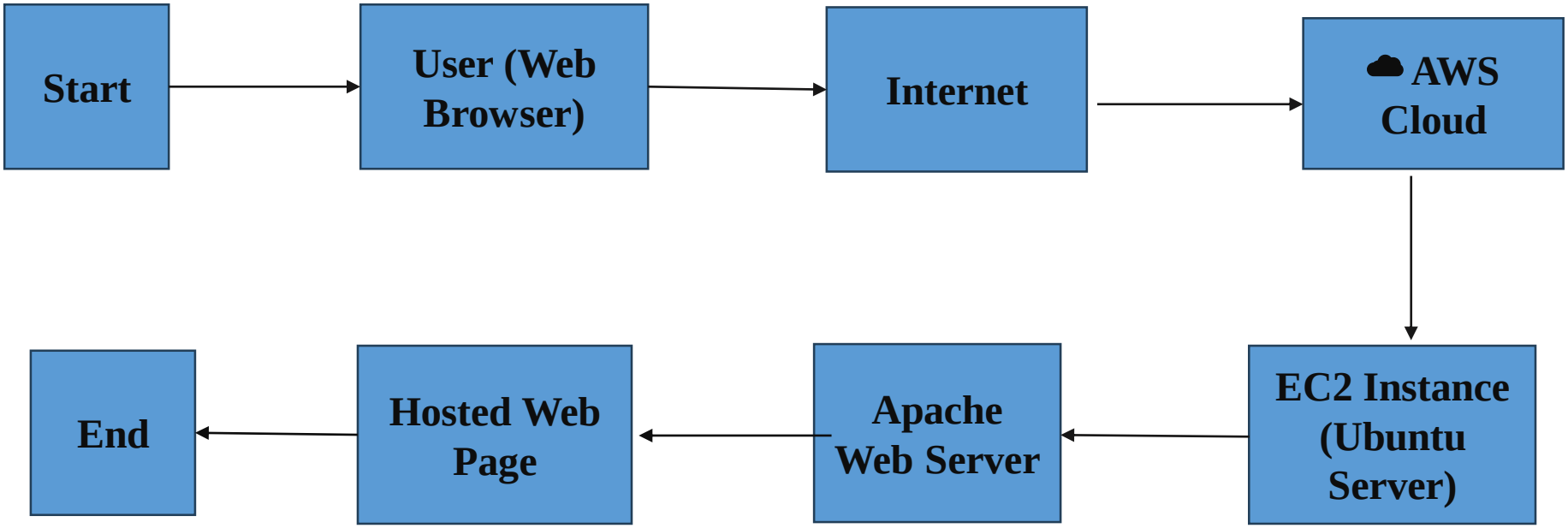
How it works

- Create an **EC2 instance** in AWS
- Install a **web server (Apache/Nginx)**
- Upload a web page
- Access it using the **public IP address**

Key Components

- AWS EC2 (Virtual Server)
- Web Server (Apache/Nginx)
- Security Groups
- SSH Key Pair

Architecture Diagram



Working Process

- User enters the **EC2 public IP address** in a web browser.
- The request travels through the **internet** to the **AWS cloud**.
- The request reaches the **EC2 virtual machine**.
- **Apache web server** processes the request.
- The **HTML webpage stored on the server** is retrieved.
- The webpage is **sent back to the user's browser** and displayed.

Real-World Example

- **Website Hosting** – Companies host their websites on cloud servers like AWS EC2.
- **E-commerce Platforms** – Online stores such as Amazon use cloud servers to handle user traffic.
- **Web Applications** – Platforms like Netflix and Spotify run their services on cloud infrastructure.
- **Company Websites** – Businesses deploy web servers in the cloud for global access.
- **Online Learning Platforms** – Educational websites use cloud servers to host content and applications.

Advantages

•**Scalability** – Resources can be increased or decreased based on demand.
•**Global Accessibility** – Websites can be accessed from anywhere through the internet.
•**Cost Effective** – Pay only for the resources used in the cloud.
•**Easy Deployment** – Web servers can be quickly created and managed.
•**High Availability** – Cloud providers ensure reliable and continuous service.
•**Secure Access Management** – Security groups control access to the server.

Conclusion

- Successfully deployed a **web server on AWS EC2**.
- Configured **security groups for secure access**.
- Installed **Apache web server** on the EC2 instance.
- Hosted a **webpage on the cloud server**.
- The website is **accessible through the public IP address**.
- This shows how **cloud computing can be used for web hosting and server management**.

References

Books

- Rajkumar Buyya – *Cloud Computing: Principles and Paradigms*
- Thomas Erl – *Cloud Computing: Concepts, Technology & Architecture*

Websites

- Amazon Web Services Official Documentation https://aws.amazon.com
- Apache HTTP Server Documentation – https://httpd.apache.org
- Ubuntu Server Guide –https://ubuntu.com/server/docs

Thank You